Sahaj Bhadja

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Summary

Software Engineer with 2+ years shipping user-facing products across **React Native**, React, and **TypeScript** clients with **real-time** sync, plus generative **media** AI with Stable Diffusion, SAM, and vision models, ramping into native iOS.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Web Technologies, Advanced NLP, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant async **AI job orchestration** with FastAPI workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Cut user-facing **dashboard latency** from 1.2s to 350ms for 500+ users by redesigning queries, indexes, and access patterns on a multi-tenant platform that held 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped a production React product integrating **vision models** (GPT-4o Vision, Claude Vision, Qwen-VL) to read handwritten work via OCR, serving 150+ instructors and 10K+ daily requests with stable latency.
- Built concurrent media ingestion for image, PDF, DOCX, and ZIP uploads with ThreadPoolExecutor batches of 1K+ files and timeout-aware model fallback, cutting feedback **turnaround** 60% at peak load.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Shipped **cross-platform** data-sharing software built as executables for Windows, Linux, and macOS, with automated VM builds, installs, and 20+ unit and integration test suites in nightly pipelines.
- Optimized API payloads for low-bandwidth clients with pagination, field filtering, and caching, improving client **response times** 30% and cutting technical debt 20% through object-oriented Singleton and Factory designs.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **React** and TypeScript interfaces with real-time WebSocket updates for resource allocation, billing, and leads, keeping 2M+ records synchronized across multiple locations with real-time allocation conflict resolution.
- Used Google Analytics on feature usage, load times, and drop-off points to **iterate** on product changes that cut resource wastage 42% and raised lead conversions 15%, with Selenium end-to-end tests.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Designed end-to-end mobile **UI/UX** flows in Figma for a cross-platform booking app, reducing design errors 25% and improving compatibility 15% across iOS and Android React Native clients.
- Built the backend for **React Native** iOS and Android clients with Firebase real-time status sync and Firebase Cloud Messaging notifications, sustaining sub-200ms latency and 40% faster booking confirmations.

Skills

Languages: TypeScript, JavaScript, Python, Java, C++, SQL
Client & Mobile: React Native, React, Next.js, Real-Time UIs, WebSockets, Firebase, Figma, UI/UX Design
Generative Media & Vision: Stable Diffusion, SAM, CLIP, BLIP, Vision Models, OCR, PyTorch, Transformers
Backend: FastAPI, Flask, Node.js, REST APIs, Async Job Orchestration, Concurrency, PostgreSQL, MongoDB
Product & Quality: Product Analytics, Iteration, Selenium, Jest, Unit Testing, CI/CD, Git
Fundamentals: Data Structures, Algorithms, OOP, Operating Systems

Projects

PiXMod Live | *Next.js, SAM, Stable Diffusion, CLIP, MongoDB, Stripe*

- Built a **generative media** editing app with SAM segmentation and object manipulation, Stable Diffusion inpainting and restoration, and CLIP search over 100K+ images, lifting premium conversion 18%.

CaptionCraft Live | *PyTorch, Transformers, BLIP, GIT, ViT-GPT-2, Gradio*

- Fine-tuned and benchmarked BLIP, GIT, and ViT-GPT-2 **vision-language** captioning models on 10K images and 100K captions, raising caption accuracy from 81% to 96% behind scalable real-time inference APIs.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented six **scheduling algorithms** in C++ with an interactive React simulator and integrated Gantt chart visualization, testing 1K+ process combinations and showing up to 25% lower average turnaround.